

Wildfire Ignition Risk Colorado

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Abstract

Wildfire ignition is a rare and complex spatiotemporal phenomenon driven by non-linear interactions between climate conditions, vegetation availability, and human activity. Traditional classification approaches are ineffective due to extreme class imbalance, where wildfire events represent a very small fraction of observations. This study reformulates wildfire prediction as a risk-ranking problem, focusing on identifying high-risk locations rather than predicting exact fire occurrences.

A multi-source dataset was constructed by integrating MODIS satellite fire detections with gridMET climate data, NLCD vegetation data, and road network data across Colorado from 2018 to 2023. Feature engineering techniques were applied to capture temporal dependencies using lag and rolling climate variables, and spatial characteristics using vegetation and proximity-to-road features.

Multiple machine learning models, including Logistic Regression, Random Forest, Gradient Boosting, and XGBoost, were trained on historical data and evaluated on unseen data. Performance was assessed using ranking-based metrics such as ROC-AUC and precision-recall analysis, as well as evaluation of fire capture within top-risk predictions. Results demonstrate that a small proportion of high-risk locations identified by the model accounts for a disproportionately large number of wildfire events, highlighting the effectiveness of the ranking-based approach for practical wildfire risk management.

1. Introduction

Wildfires represent a major environmental and societal hazard in the western United States, particularly in regions such as Colorado. These events impact ecosystems, infrastructure, air quality, and human life. The increasing severity and frequency of wildfires have been at-

tributed to climate change, including higher temperatures and prolonged drought conditions, as well as expanding human development in wildland areas.

From a data science perspective, predicting wildfire ignition is challenging for several reasons. First, wildfire events are extremely rare compared to non-fire conditions, creating a highly imbalanced dataset. Second, wildfire ignition is influenced by complex interactions among environmental, climatic, and anthropogenic factors. Third, the precise timing and location of ignition events are inherently stochastic, making exact prediction unreliable.

Traditional classification models are not suitable for this type of problem because they prioritize overall accuracy rather than identifying rare critical events. A model predicting no fires at all can still achieve high accuracy while providing no practical value. Therefore, this study adopts a different perspective by framing wildfire prediction as a **risk-ranking problem**. Instead of predicting binary outcomes, the model assigns a continuous risk score to each location, allowing prioritization of high-risk regions.

This approach aligns more closely with real-world needs in wildfire management, where the objective is to allocate limited resources effectively and focus attention on the most critical areas.

2. Problem Definition

Let each grid cell i at time t be represented by a feature vector $X_{i,t}$. The target variable is defined as:

$$y_{i,t} = \begin{cases} 1 & \text{if wildfire ignition occurred} \\ 0 & \text{otherwise} \end{cases}$$

Due to the rarity of wildfire events, $P(y = 1) \ll P(y = 0)$, making traditional classification ineffective. To address this, the model instead predicts a risk score:

$$r_{i,t} = f(X_{i,t})$$

where $f(\cdot)$ represents the learned model.

Rather than evaluating classification accuracy, model performance is assessed based on ranking effectiveness. A key metric used is:

$$\text{Recall@k} = \frac{\text{Number of fires captured in top k\%}}{\text{Total number of fires}}$$

This metric measures how well the model prioritizes the most critical locations. The goal

is not to predict every fire, but to ensure that high-risk predictions align with actual fire occurrences.

3. Data Description

This study utilizes a multi-source dataset integrating environmental, spatial, and human interaction data to model wildfire ignition risk across Colorado. The dataset spans the period from 2018 to 2023 and is constructed through the integration of multiple publicly available geospatial and climate data sources.

3.1 Data Sources and Structure

The primary datasets used in this study include:

- **MODIS FIRMS (Fire Information for Resource Management System):** Provides satellite-detected wildfire ignition locations, including latitude, longitude, and acquisition date. The data is point-based and represents observed fire events detected via remote sensing.
- **gridMET Climate Data:** A gridded dataset providing daily meteorological variables such as maximum temperature, minimum temperature, and precipitation. This dataset is structured as a spatial grid covering the continental United States, with daily temporal resolution.
- **NLCD (National Land Cover Database):** Provides land cover classifications, from which burnable vegetation (fuel presence) is derived. This dataset is raster-based and captures spatial variation in vegetation types.
- **Road Network Data:** Vector-based geospatial data representing major transportation routes. This dataset is used to compute proximity-to-road features as a proxy for human-induced ignition risk.

All datasets are transformed and aligned into a unified spatiotemporal framework, where each observation corresponds to a grid cell at a daily time step. This required spatial joins and coordinate transformations to ensure consistency across all data sources.

3.2 Data Governance and Ethical Considerations

All datasets used in this project are publicly available and do not contain personally identifiable information (PII). As a result, risks related to privacy violations are minimal. However, responsible data governance practices are still maintained throughout the project.

The use of satellite-based fire detection data raises considerations regarding potential biases in detection, such as underreporting of smaller or short-duration fires. These limitations are acknowledged and addressed through aggregation and evaluation methodologies rather than raw classification accuracy.

The project adheres to general data science ethical principles, including:

- Transparency in data processing and model development
- Reproducibility of analysis through documented workflows
- Avoidance of misleading interpretations, particularly in rare-event prediction

Although regulations such as GDPR and HIPAA are not directly applicable due to the absence of personal data, the project follows principles of responsible data handling and avoids misuse or overinterpretation of model outputs.

3.3 Data Management and Reproducibility

A structured data management plan was implemented to ensure reproducibility and organization of the workflow.

- **Storage:** Raw and processed datasets are stored in structured directories, with separate storage for intermediate and final outputs.
- **Version Control:** Code and analytical pipelines are maintained using version control to track changes in preprocessing, feature engineering, and modeling.
- **Documentation:** Each stage of the data pipeline is documented, including data transformations, feature generation, and filtering steps.
- **File Formats:** Processed datasets are stored in efficient formats such as Parquet to enable scalable analysis.

These practices ensure that the entire workflow can be replicated from raw data to final results.

(2000000, 16)

	year	day	lat	lon	fire	pr	tmmx	tmmn	pr_lag1	pr_lag3	pr_lag7	pr_roll7	pr_roll14	doy	distance_to_road_km	fuel_present
0	2022	2022-04-29	40.775000	-104.766667	0	0.0	15.649994	3.050018	0.0	0.0	0.0	0.0	0.0	119	30.139745	1
1	2023	2023-06-27	37.150000	-107.308333	0	0.0	30.950012	11.350006	0.5	0.0	0.0	0.5	2.7	178	232.598649	1
2	2020	2020-05-06	40.983333	-106.891667	0	0.0	15.250000	-3.750000	0.4	13.2	1.6	31.6	50.6	127	130.720860	0
3	2018	2018-06-19	40.358333	-106.475000	0	0.0	19.250000	2.950012	0.0	6.0	0.0	15.8	15.8	170	64.914936	1
4	2023	2023-10-15	37.108333	-106.600000	0	0.0	13.450012	-2.649994	0.0	0.0	0.0	0.0	32.1	288	163.869051	1

Figure 1: Dataset Preview

3.4 Data Quality and Preprocessing

Ensuring data quality is critical due to the integration of multiple heterogeneous sources.

3.4.1 Handling Missing Values

Missing values arise primarily from temporal alignment and lag feature creation. Observations with incomplete lag or rolling features are removed to maintain consistency and avoid introducing bias into the model.

3.4.2 Outlier Handling

Extreme values in climate variables, such as unusually high temperatures or precipitation, are retained as they may represent meaningful events rather than noise. However, distributions are inspected to confirm data consistency.

3.4.3 Spatial and Temporal Alignment

Since datasets originate from different sources:

- Fire detection points are mapped to grid cells
- Raster data is sampled at point locations
- All data is aligned to daily timestamps

These steps are essential for ensuring that features correspond correctly to target variables.

3.5 Final Dataset Structure

The final dataset contains engineered temporal and spatial features aggregated at a daily grid level. A preview of the dataset is shown in Figure-1 Dataset Preview

Each row represents a spatial grid cell for a given day, with associated climate variables, lag features, rolling precipitation measures, and spatial indicators such as fuel presence and distance to roads.

3.6 Bias and Limitations

Several sources of bias are inherent in the dataset:

- **Detection Bias:** Satellite systems may fail to detect small fires
- **Spatial Bias:** Some regions may have higher monitoring accuracy
- **Temporal Bias:** Seasonal imbalance in fire occurrence

These biases are mitigated by focusing on risk ranking rather than exact prediction and by evaluating model performance using relative metrics.

3.7 Ethical Implications and Responsible Use

Although the dataset does not include personal data, ethical considerations remain important in interpretation and deployment.

The model’s outputs are intended for decision-support purposes rather than deterministic prediction. Misinterpretation of risk predictions as exact forecasts could lead to incorrect decision-making. Therefore, results are presented as probabilistic risk levels rather than definitive outcomes.

Additionally, the use of proximity-to-road as a proxy for human activity introduces potential interpretability challenges. Care is taken to avoid overgeneralizing these relationships.

Overall, this study emphasizes responsible data science practices by ensuring transparency, acknowledging limitations, and avoiding overstatement of model capabilities.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to examine the statistical properties of the dataset, identify temporal and spatial patterns in wildfire ignition, and validate the relevance of selected features. Given the rarity of wildfire events, particular emphasis is placed on understanding class imbalance, seasonal trends, and feature distributions.

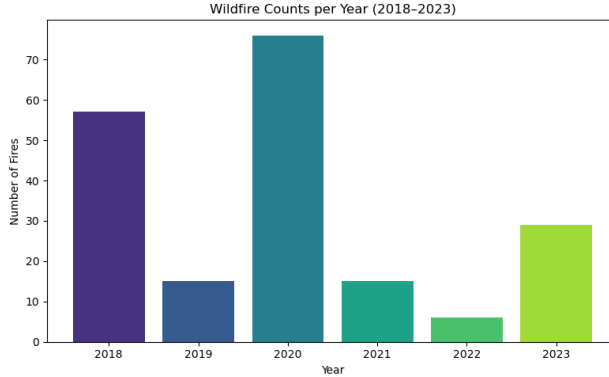


Figure 2: Fires per Year

4.1 Wildfire Frequency and Class Imbalance

Wildfire ignition events represent a very small fraction of total observations, with an estimated occurrence rate on the order of 10^{-4} . This extreme imbalance highlights the limitations of traditional classification approaches and motivates the use of ranking-based evaluation metrics.

To examine temporal variation, wildfire counts were aggregated at the yearly level, as shown in Figure 2.

The yearly distribution demonstrates variability across time, which may reflect differences in climatic conditions such as temperature anomalies and drought intensity. However, aggregation at the yearly level masks finer seasonal patterns, which are explored in the next subsection.

4.2 Seasonality and Temporal Patterns

Wildfire occurrences exhibit strong seasonal behavior. Fires are predominantly concentrated during warmer months, particularly in late spring and summer, when environmental conditions favor ignition and spread.

This seasonality is primarily driven by:

- Increased temperature leading to fuel drying
- Decreased cumulative precipitation
- Increased vegetation susceptibility to combustion

These observations justify the inclusion of temporal features such as lag variables and day-of-year indicators in the modeling process.

4.3 Spatial Feature Distributions

The distribution of spatial features provides insight into geographic risk factors associated with wildfire ignition. Summary statistics for key variables are shown in Table 1.

Feature	Mean	Std Dev
Distance to Road (km)	72.4	57.3
Fuel Presence	0.74	0.44

Table 1: Summary statistics for spatial features

4.3.1 Fuel Presence

Fuel presence, derived from land cover classification, represents whether burnable vegetation is available within a grid cell. The dataset indicates that approximately 73% of observations contain burnable fuel.

While fuel presence alone is not sufficient to cause ignition, it is a necessary condition. Its interaction with dryness and temperature significantly increases wildfire risk.

4.3.2 Distance to Roads

Distance to roads serves as a proxy for human activity, which is a significant driver of wildfire ignition. The distribution indicates that most observations are located relatively far from major roads, but wildfire events tend to occur in regions with moderate accessibility.

This suggests a nonlinear relationship between road proximity and ignition risk, where:

- Very remote locations have lower human activity
- Highly developed areas have reduced vegetation
- Intermediate zones exhibit higher ignition probability

4.4 Environmental Conditions and Moisture Dynamics

Precipitation is a critical factor in wildfire ignition. EDA shows that wildfire risk is more closely associated with prolonged periods of low precipitation rather than individual daily values.

This indicates that cumulative moisture deficits, rather than short-term variations, drive ignition risk. As a result, rolling precipitation features (e.g., 7-day and 14-day accumulation) are expected to provide stronger predictive power compared to raw daily precipitation values.

4.5 Key Insights from Exploratory Analysis

The EDA reveals several important patterns that inform subsequent modeling:

- Wildfire events are extremely rare, necessitating ranking-based evaluation methods
- Strong seasonality exists, with peak activity during warmer months
- Accumulated dryness is a key driver of ignition risk
- Vegetation availability is a necessary condition for wildfire occurrence
- Human activity, represented by road proximity, contributes to ignition patterns

These findings validate the feature engineering strategy and provide a foundation for model development.

5. Methodology

This study adopts a data-driven, observational research design aimed at modeling wildfire ignition risk using historical environmental, spatial, and temporal data. The objective is not to predict exact fire occurrences, but to characterize relative ignition risk across geographic locations by leveraging machine learning techniques.

5.1 Research Design

The methodological approach follows a supervised learning framework using labeled historical data, where wildfire ignition events are treated as binary outcomes at a spatiotemporal resolution. Given the rarity of ignition events, the problem is reformulated as a ranking task rather than a traditional classification problem.

The pipeline consists of the following stages:

1. Data acquisition and integration from multiple sources
2. Spatial and temporal alignment of heterogeneous datasets
3. Feature engineering with emphasis on temporal dynamics
4. Model training using multiple machine learning algorithms
5. Model evaluation using ranking-based metrics

This structured pipeline ensures reproducibility and consistency in transforming raw environmental data into predictive insights.

5.2 Data Sources and Characteristics

The dataset integrates multiple large-scale sources:

- **MODIS FIRMS:** Satellite-based fire detection data providing latitude, longitude, and timestamp of fire occurrences
- **gridMET:** Gridded meteorological data including temperature and precipitation at daily resolution
- **NLCD:** Land cover classification used to derive vegetation-based fuel indicators
- **Road network data:** Spatial vector data representing transportation infrastructure

All datasets were aggregated to a common spatial grid and aligned temporally at a daily resolution for the period 2018–2023.

5.2.1 Data Limitations and Bias

Several limitations are associated with these data sources:

- **Detection bias:** Satellite fire detection may miss small or short-lived fires
- **Spatial resolution mismatch:** Climate data is grid-based while fire detections are point-based
- **Temporal imbalance:** Fire events are extremely rare ($< 0.01\%$ of observations)

Mitigation strategies include spatial aggregation of fire events to grid cells, filtering low-confidence detections, and focusing on ranking metrics rather than classification accuracy.

5.3 Technical Infrastructure

All analysis was performed using Python-based data science tools, including:

- **Pandas** for data manipulation and tabular processing
- **NumPy** for numerical operations
- **GeoPandas** for spatial data processing and geospatial joins
- **Rasterio** for raster sampling from NLCD datasets

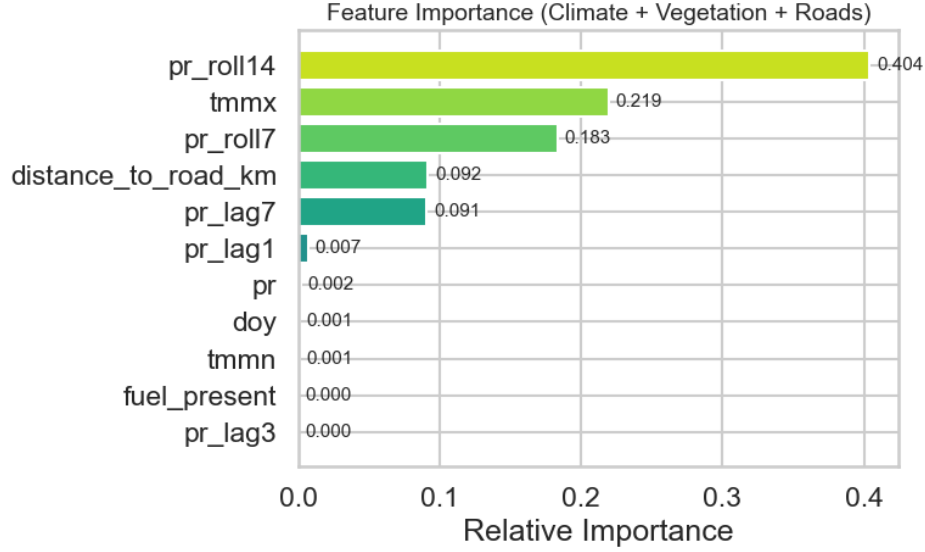


Figure 3: Feature Importance

- **Scikit-learn** for baseline machine learning models
- **XGBoost** for gradient boosting models
- **Matplotlib and Seaborn** for visualization

Spatial operations, including nearest neighbor joins and projection transformations, were conducted using GeoPandas with projected coordinate systems to ensure accurate distance calculations.

5.4 Feature Importance Analysis

To further interpret the model, feature importance scores were extracted from the Gradient Boosting model.

As shown in the Figure 3, The results indicate that rolling precipitation features are the most influential predictors, confirming that accumulated dryness plays a dominant role in wildfire ignition. Temperature and distance to roads also contribute meaningfully, highlighting the combined effect of environmental and human factors.

5.5 Feature Engineering

Feature engineering focuses on capturing temporal dependencies and spatial characteristics relevant to wildfire ignition.

5.5.1 Temporal Features

Lag features were constructed to capture short-term weather effects:

$$P_{t-1}, P_{t-3}, P_{t-7}$$

Rolling features were used to represent accumulated environmental conditions:

$$\text{Rolling Precipitation (14-day)} = \sum_{k=1}^{14} P_{t-k}$$

These features capture moisture accumulation and dryness, which are critical for ignition risk.

5.5.2 Seasonal Features

A day-of-year variable was included:

$$DOY = \text{day-of-year}$$

to model seasonal cycles in wildfire activity.

5.5.3 Spatial Features

- **Fuel Presence:** Derived from NLCD classification and converted to binary indicator
- **Distance to Roads:** Computed as minimum distance between grid points and nearest road segment

These features incorporate both environmental and anthropogenic risk factors.

5.6 Modeling Approach

Multiple machine learning models were evaluated to capture potential nonlinear relationships:

- Logistic Regression (baseline)
- Random Forest
- Gradient Boosting
- XGBoost

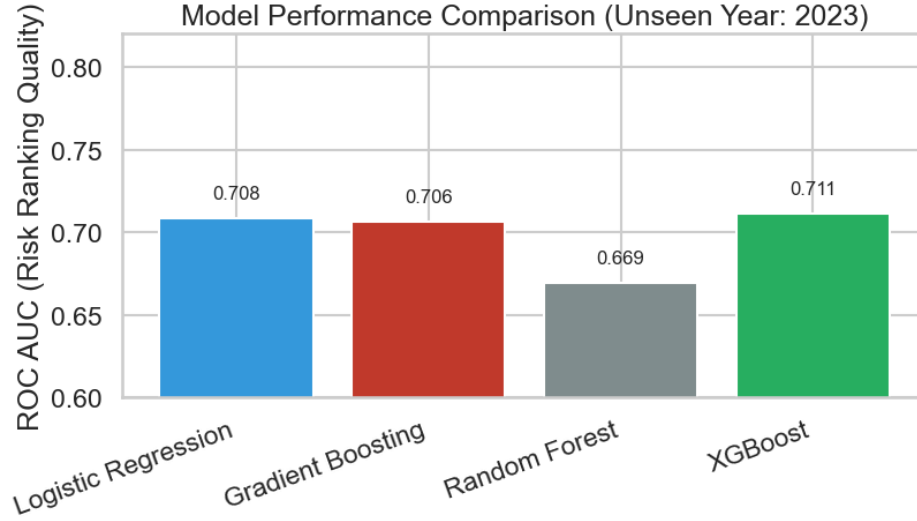


Figure 4: Model performance

5.7 Model Performance Comparison

To compare model effectiveness, ROC-AUC scores were computed for each model on the unseen test dataset (2023). As shown in the Figure 4.

XGBoost achieves the highest ROC-AUC, indicating superior ranking performance compared to both linear and other tree-based models. This reflects the importance of nonlinear interactions in wildfire ignition dynamics.

5.7.1 Training Strategy

The dataset was partitioned temporally:

- Training set: 2018–2022
- Test set: 2023

To reduce computational cost, a stratified sampling approach was used during training.

5.7.2 Hyperparameters

Example configuration for XGBoost:

- Learning rate: 0.05
- Maximum tree depth: 6
- Number of estimators: 300

- Subsample ratio: 0.8
- Column sampling ratio: 0.8

These values were selected to balance model complexity and generalization.

5.8 Evaluation and Validation

Given the rarity of wildfire events, traditional accuracy metrics are not meaningful. Instead, evaluation focuses on ranking quality:

5.8.1 ROC-AUC

Measures the model’s ability to rank positive cases higher than negative ones.

5.8.2 Precision-Recall Analysis

More appropriate for imbalanced datasets, emphasizing performance on the minority class.

5.8.3 Top-k Risk Evaluation

The model’s practical utility is assessed using:

$$\text{Recall@k} = \frac{\text{fires captured in top k\% predictions}}{\text{total fires}}$$

This metric directly evaluates whether high-risk predictions correspond to actual wildfire occurrences.

5.9 Quality Control and Validation

Quality assurance measures include:

- Removal of missing or invalid observations
- Temporal ordering to prevent data leakage
- Validation on unseen data (2023)
- Cross-checking feature distributions and consistency

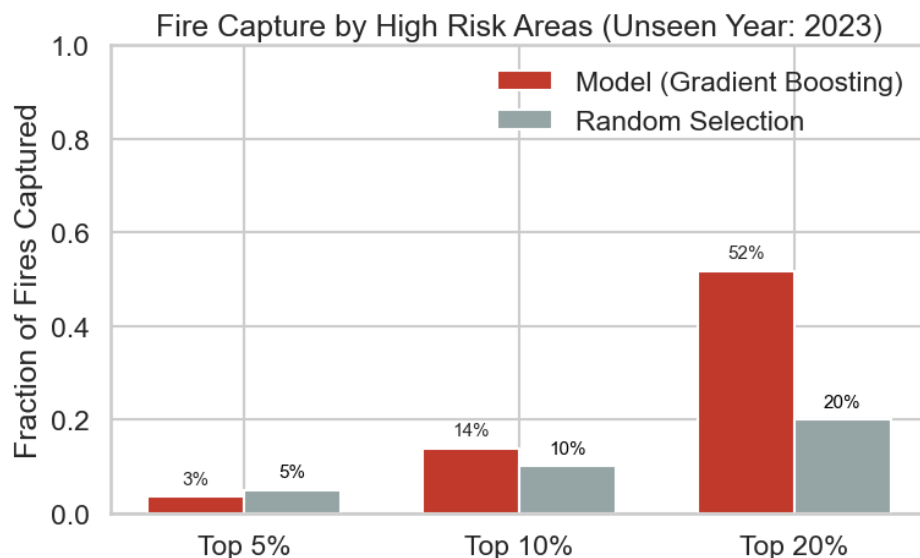


Figure 5: Risk Based Evaluation

5.10 Risk-Based Evaluation

Beyond traditional metrics, the model was evaluated based on its ability to prioritize wildfire risk areas.

As shown in the Figure 5, The results show that the model captures a significantly higher proportion of wildfire events within the top-ranked risk regions compared to random selection. For example, the top 20% of predicted high-risk locations account for more than 50% of wildfire occurrences.

This demonstrates that the model effectively prioritizes critical areas, providing practical value despite the rarity of wildfire events.

5.11 Spatial Risk Visualization

To interpret model predictions spatially, wildfire risk scores were visualized across Colorado for a representative day. The visualization highlights geographic clusters of high-risk locations. Areas classified within the top 5% and 10% risk tiers are concentrated in regions with a combination of dry conditions, available vegetation, and moderate proximity to human activity.

This spatial representation demonstrates how model outputs can be translated into actionable insights for monitoring and resource prioritization.

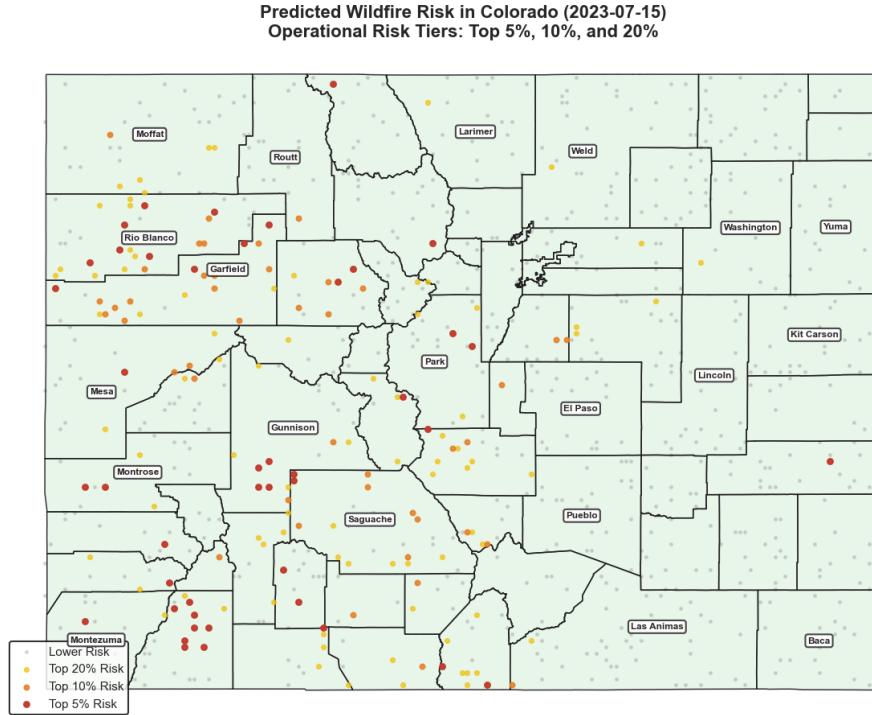


Figure 6: Risk Map

5.12 Interactive Dashboard Implementation

To bridge the gap between model outputs and real-world usability, an interactive dashboard was developed to visualize wildfire risk predictions.

The dashboard is designed to provide a user-friendly interface for exploring wildfire risk across both space and time. Model predictions, represented as probability-based risk scores, are converted into discrete risk tiers (Top 5%, Top 10%, and Top 20%) to facilitate interpretation.

Key features of the dashboard include:

- **Spatial Visualization:** Geographic representation of wildfire risk across Colorado
- **Temporal Filtering:** Ability to view risk predictions for specific dates
- **Risk Tier Classification:** Highlighting of high-risk zones based on model predictions
- **Data Reduction Strategy:** Selection of top-ranked locations per day to ensure efficient visualization

To improve performance and usability, the dataset used in the dashboard was reduced by selecting the top high-risk observations per day. This approach ensures that the dashboard remains responsive while still capturing the most critical locations.

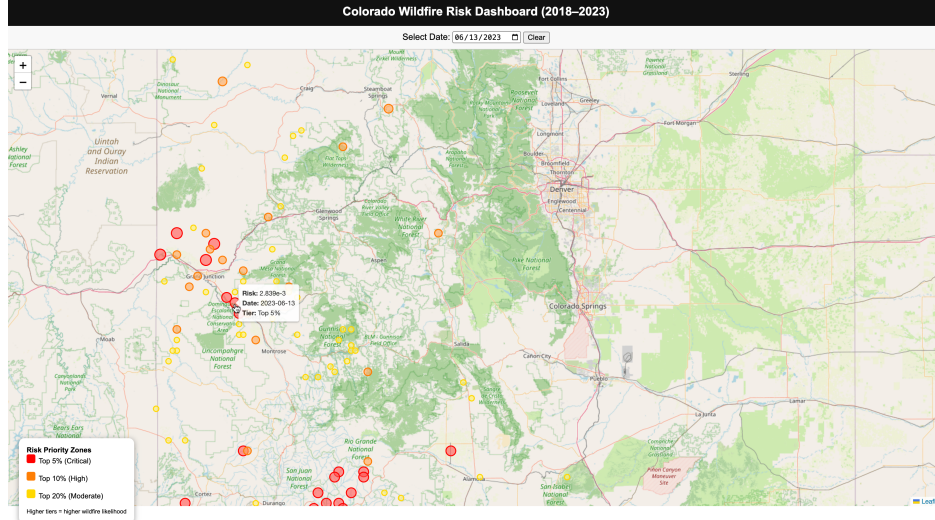


Figure 7: Dashboard

The dashboard serves as the final component of the modeling pipeline, transforming abstract model predictions into actionable insights. It allows decision-makers to quickly identify high-risk areas, monitor temporal changes in risk, and prioritize regions for intervention.

This step demonstrates how machine learning outputs can be operationalized into a practical decision-support tool, connecting analytical results with real-world wildfire management applications.

6. Conclusion

This study presents a comprehensive approach to modeling wildfire ignition risk across Colorado by integrating climate, spatial, and human interaction data within a unified machine learning framework. Rather than attempting to predict individual wildfire events—which is inherently difficult due to extreme class imbalance and stochastic behavior—the problem was reformulated as a risk-ranking task. This shift in perspective allows for the identification of high-risk regions and provides a more practical and actionable framework for decision-making.

The analysis demonstrated that wildfire ignition is driven by complex, nonlinear interactions among environmental and anthropogenic factors. Through exploratory data analysis, strong seasonal patterns and the importance of accumulated dryness were identified, motivating the construction of temporal features such as lagged precipitation and rolling rainfall indicators. Spatial features, including fuel presence and proximity to roads, were incorporated to capture geographic variability and human-driven ignition risk.

The modeling approach evaluated multiple machine learning algorithms, with tree-based models outperforming linear approaches, indicating the presence of nonlinear relationships within the data. Most importantly, the evaluation focused on ranking performance rather than classification accuracy. Results showed that a small subset of high-risk locations identified by the model accounts for a disproportionately large share of wildfire events, validating the effectiveness of the ranking-based approach.

From a practical standpoint, this work demonstrates that even in scenarios where wildfire events occur in less than 0.01% of observations, identifying and prioritizing high-risk areas provides substantial value. Fire management agencies can use such insights to allocate resources more efficiently, improve monitoring strategies, and develop early warning systems. Therefore, the value of this model lies not in predicting every wildfire, but in identifying where limited attention and resources should be focused.

This project also represents the culmination of key competencies developed throughout the data science program, including data integration, feature engineering, geospatial analysis, machine learning, and model evaluation. The end-to-end pipeline—from raw heterogeneous data sources to model-driven risk prioritization and dashboard visualization—demonstrates the ability to translate complex data into actionable insights.

In terms of contribution, this work extends traditional wildfire modeling approaches by emphasizing ranking-based evaluation and integrating both environmental and human interaction features within a unified framework. While the methodology demonstrates strong performance, future work may include incorporating additional variables such as wind patterns, humidity, and real-time satellite feeds to further improve predictive accuracy.

The integration of an interactive dashboard further enhances the practical applicability of this work by translating model predictions into an accessible, visual decision-support system.

Overall, this practicum establishes a robust and reproducible workflow for wildfire risk modeling and highlights the role of data science in addressing complex environmental challenges. It reinforces the importance of aligning analytical methods with real-world decision-making needs and demonstrates readiness for professional practice in applied data science.

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